Song Database Management Program User Guide

Goal of the Program

The Song Database Management Program is designed to provide a user-friendly interface for managing a database of songs. The program allows users to perform various operations, including adding new song entries, listing all entries, searching for specific songs, deleting songs, and saving the database to a file. The primary goal is to facilitate efficient organization and retrieval of information related to a collection of songs.

Instructions on How to Use the Program

1. Getting Started

Compile the Program: Ensure that you have a C compiler installed on your system. Compile the program using the provided source code.

Run the Program: Execute the compiled program to launch the Song Database Management interface.

2. Main Menu

Upon running the program, you will be presented with a main menu containing several options:

Add new entry to database: Allows you to add new songs to the database.

List all entries of the database: Displays a list of all songs in the database.

Search and display songs: Enables searching for songs based on keywords.

Delete any song from the database: Lets you delete a specific song from the database.

Exit: Saves the current state of the database and exits the program.

3. Adding New Songs

- Choose option 1 from the main menu.
- Enter information for each song, including title, artist, album, release year, genre, and song length (in minutes and seconds).

- Repeat the process to add multiple songs by entering ‘Y’ or ‘y’.

4. Listing All Songs

- Choose option 2 from the main menu.
- View a comprehensive list of all songs in the database, including details such as title, artist, album, release year, genre, and song length.

5. Searching for Songs

- Choose option 3 from the main menu.
- Enter a keyword (e.g., title, artist, album, or genre) to search for specific songs.
- View matching results.

6. Deleting Songs

- Choose option 4 from the main menu.
- Enter the title of the song you want to delete.
- Confirm the deletion when prompted.

7. Exiting the Program

- Choose option 5 from the main menu.
- The program will save the current state of the database to a file named "song_database.txt" and free allocated memory before exiting.

File Operations

Saving and Loading the Database

- The program automatically loads the song database from the file "song_database.txt" upon startup, creating an empty database if the file doesn't exist.
- When exiting the program (choosing option 5), the current state of the database is saved to the same "song_database.txt" file.

Follow these instructions to efficiently manage your song database using the provided Song Database Management Program.